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Cysteinyl leukotriene signaling promotes burn wound healing

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Cysteinyl leukotrienes (CysLTs; LTC_4, LTD_4, and LTE_4) are inflammatory mediators known for their involvement in bronchoconstriction, asthma and allergy, and primarily signal through the receptors CysLT1 and CysLT2. Interestingly, recent studies have found that CysLT receptors are expressed in normal skin, and that CysLT signaling may interfere with wound healing. Furthermore, our preliminary data show that enzymes associated with CysLT synthesis are highly upregulated in burned murine skin compared to healthy skin, while a previous study showed that blister fluids from burn patients contain high LTC_4 levels. Collectively, these observations suggest that CysLTs may propagate inflammation at the burn injury site. Conversely, previous *in vitro* studies demonstrate that CysLT signaling promotes contraction and collagen production by fibroblasts, and induces proliferation of endothelial cells, suggesting that CysLTs may promote healing after injury. To elucidate the effects of CysLTs in an inflammatory wound such as burn, we induced scald burn injury to C57BL/6 mice treated with montelukast, a FDA-approved CysLT1 antagonist, or vehicle. Intriguingly, we found that montelukast-treated mice exhibited significantly delayed re-epithelialization, which was associated with fewer proliferating keratinocytes and endothelial cells, when compared to the vehicle-treated control mice. We also demonstrate that montelukast-treated wounds expressed lower levels of pro-healing genes, and higher levels of the inflammatory gene, *inos*. Collectively, our data identify a new mechanism activated at the burn wound site, and strongly suggest that CysLT signaling promotes burn wound healing. This study reveals a novel therapeutic potential of exploiting the CysLT1 signaling pathway to accelerate healing in burn patients.

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Mechanobiological study of keloid: from bedside to bench

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Keloids are pathological scars prone to form in body sites with increased skin tension. Caveolin-1 (CAV1) has been associated with the regulation of cell mechanics, including cell softening and loss of stiffness sensing ability. Although CAV1 is present in low amounts in keloid fibroblasts (KFs), the causal association between CAV1 downregulation and its aberrant responses to mechanical stimuli remain unclear. In this study, atomic force microscopy showed that KFs were softer than normal fibroblasts with a loss of stiffness sensing. The decrease of CAV1 contributed to the hyperactivation of fibrogenesis-associated RUNX2, a transcription factor germane to osteogenesis/chondrogenesis, and increased migratory ability in KFs. Treatment of KFs with trichostatin A (TSA), which increased the acetylation level of histone H3, increased CAV1 and decreased RUNX2 and fibronectin. TSA treatment also resulted in cell stiffening and decreased migratory ability in KFs. Collectively, these results suggest a novel role for CAV1 downregulation in linking the aberrant responsiveness to mechanical stimulation and extracellular matrix accumulation with the progression of keloids, findings that may lead to new developments in the prevention and treatment of keloid scarring.

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BPM31510 improves wound healing and skin integrity in epidermolysis bullosa patients

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Epidermolysis bullosa (EB) is a genetic mechanobullous disorder characterized by skin fragility susceptible to blistering with minimal trauma due to malfunction or lack of anchoring proteins within different levels of the basement membrane zone. EB patients often experience chronic wounds due to multiple factors such as prolonged inflammatory phase, infection, nutrition, tissue oxygenation and medications (corticosteroids). BPM315103.0% is a Coenzyme Q10 containing cream that has shown to increase the expression of structural proteins (collagens, plectin, laminins, KRT13, KRT14 and KRT17) in fibroblasts and keratinocytes. We hypothesize that topical BPM315103.0% improves wound healing and skin fragility in EB patients. We evaluated 4 patients with simplex, junctional and dystrophic EB, aged 16 - 60. CelluTome Epidermal Harvesting System was used to create even wounds of 1.8 mm by suctioning the epidermal layer of unaffected areas of the skin. Wounds were created before (day 1 week 1) and after (day 1 week 8) treatment with topical BPM315103.0%. We performed *in vivo* Reflectance Confocal Microscopy (RCM) evaluation immediately, 4 hours and 8 hours after the suction wounds by CelluTome were created on week 1 and 8. RCM allowed us to evaluate wound healing response noninvasively, at the cellular-resolution. RCM images showed more elongated keratinocytes and few inflammatory cells at the wound bed on week 8 in all 4 patients. We also observed thicker and more organized collagen bundles and more fibroblasts on week 8 of the wound base. In addition, well-defined wounds were created on week 8 for all 4 patients. No adverse reaction was reported thus far. In this small set of EB patients, we found BPM315103.0% to be well tolerated and additionally their intact skin demonstrated improved re-epithelization, with thicker and more organized collagen bundles and fibroblasts after 8 weeks of application.

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Tissue mechanics partake in spatiotemporal patterning of wound-induced hair neogenesis in African spiny mouse

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The African spiny mouse (Acomys) can regenerate up to 70% of its skin, including all appendages, after full-thickness skin wounding. This wound-induced hair neogenesis (WIHN) begins from the wound periphery on post-wound day 15 (PWD15), and later in the wound center (PWD21). In contrast, WIHN only occurs in the center of the wound for C57Bl/6 mouse. The spatiotemporal stiffness of the spiny mouse wounds was measured using the atomic force microscopy, which showed a gradual increase from 2.35 kPa (wound center) and 7.87 kPa (wound periphery) on PWD15 to 7.19 kPa and 11.45 kPa on PWD21, respectively. These results suggest there is an optimal range of tissue stiffness at 7-12 kPa for WIHN to occur in spiny mouse. Time course RNA-seq results show that epithelial-to-mesenchymal transition and inflammation-related genes were upregulated in the wounds of spiny mice on PWD15; however, Runx3 and extracellular matrix (ECM) genes were not elevated. The soft mechanical characteristics and high collagen III to I ratio ECM expression profile of the spiny mouse wounds are similar to that of fetal skin. We speculate that spiny mouse builds a relatively lower stiffness skin which breaks easily to provide an opportunity to escape from predators, and its wound heals softly to dampens the inflammatory response during wound healing, fostering the optimal mechanical environment for skin regeneration.

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A synthetic, flowable wound matrix that improves vascularization and modulates immune response in porcine wound healing

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Chronic wounds affect over 5 million patients and costing an estimated $20 billion annually, representing a significant burden to patients and the US health care system. Naturally sourced cellular and tissue/growth factor therapies exist to treat chronic wounds but their cost prevents reimbursed use until wounds are chronic, decreasing effectiveness and hindering outcome. Cost-effective, simple treatments are needed that can accelerate healing in acute wounds and prevent development of chronic wounds. To address this need, we have developed a new wound matrix technology - the microporous annealed particle (MAP) scaffold. The MAP approach is completely free of biologics and cells - focusing on unique physical geometries in synthetic flowable materials to elicit regenerative responses. This is achieved through the use of concentrated solutions of polyethylene glycol micro-hydrogels that, once delivered to the wound bed, 'anneal' to one another and the surrounding tissue, forming a hyper-porous scaffold between the microgels that promotes accelerated healing and vascularization. In this work, we tested if MAP can improve healing in a porcine excisional wound model, compared to Aquafor (moist wound healing) and the leading decellularized matrix - Oasis SIS wound matrix (Smith&Nephew). After 5 days, MAP better filled the wound defect compared to Oasis or Aquafor, promoted deep vascularization, and decreased acute inflammation. After 14 days, both Oasis and MAP promoted larger caliber, more mature blood vessels compared to Aquafor treated wounds, increased tissue thickness, and decreased atrophy. Further, MAP formulation variants were capable of controlling tissue immune state from below baseline ($\sim$75%) to 2x increase compared to Aquafor. These findings demonstrate that the MAP gel improves vascularization and modulates wound immune state without growth factors or cells, showing great promise as an acute wound treatment to mitigate chronic wound progression.

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Topical beta-blockers and wound healing

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Some chronic wounds fail to heal despite standard therapy. Recently, the armamentarium available to treat chronic wounds has been enriched with the newly discovered functions of beta-blockers (BB). BB are relatively safe and been in use for decades for various indications and therefore represent exciting potential for translation to the bedside. Herein we present clinical outcomes to support existing in vitro evidence for use of BB for wound epithelialization. Three patients were treated with topical timolol maleate 0.5% at a 1 drop/cm2 dose. The frequency of application hinged on dressing changes, ranging from daily to biweekly. At baseline the average wound area was 41.8 cm2. Two patients had venous leg ulcers with significantly different confounding comorbidities: one had history of acute myeloid leukemia, chemotherapy and bone marrow transplant, and the other had erosive pustular dermatosis of the leg. The third patient had a chronic abdominal ulcer secondary to surgical bowel resection due to complications of ventral hernia. All patients received standard care relevant to their wound etiology. All wounds appeared unable to epithelialize. Use of timolol induced full epithelialization in all wounds after 3-8 months of treatment. In our series, BBs were used successfully to enhance epithelialization. Beta 2 adrenergic receptors are expressed on membranes of keratinocytes, fibroblasts, and melanocytes. Their blockade interferes with mitogen-activated protein kinase signaling inducing cell migration, proliferation, re-epithelialization, contraction and angiogenesis. Our anecdotal success in healing wounds across variable etiologies suggests a canonical healing pathway modulated by BB. More evidence for the role of BB in healing wounds of specific etiology is needed. Patients may benefit from use of this inexpensive, simple and readily available therapy for chronic wounds.